Deliverable #1: Software Requirement Specification (SRS)

SE 3A04: Software Design II – Large System Design

Tutorial Number: T02
Group Number: G1
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1 Introduction

This Software Requirements Specification (SRS) document describes the requirements for the Smart City Environmental Monitoring & Alert System (**SCEMAS**), a cloud-native, **IoT**-based software platform designed to collect, process, and analyze environmental data from a distributed network of sensors deployed across an urban environment. The system aims to transform real-time sensor measurements into actionable information for city officials and the general public through automated alerting, visualization, and data access services.

1.1 Purpose

The purpose of this SRS is to formally specify the functional and non-functional requirements of the Smart City Environmental Monitoring & Alert System (**SCEMAS**). It defines what the system must do, the conditions under which it must operate, and the quality attributes it must satisfy, without prescribing specific implementation details.

The intended audience of this document includes the student development team, who will use it as the primary reference for design and implementation, as well as any stakeholders acting in the role of city operators, administrators, or public users, whose needs are represented by the system requirements.

1.2 Scope

The software product specified in this document is the Smart City Environmental Monitoring & Alert System (**SCEMAS**).

SCEMAS is a cloud-based **IoT** platform that ingests real-time telemetry from simulated environmental sensors using standard protocols (such as **MQTT**). It validates, stores, and processes time-series environmental data (e.g., air quality, temperature, humidity, and noise levels), detects threshold violations and anomalous patterns using a rule-based alerting engine, and generates and manages alerts, including logging, acknowledgment, and resolution.

The platform also provides a secure web dashboard for city operators to visualize sensor data, alerts, and system health, exposes a public read-only REST API for accessing aggregated non-sensitive environmental information, and enforces security, authentication, authorization, auditing, and privacy-by-design principles.

1.3 Definitions, Acronyms, and Abbreviations

- **IoT**: Internet of Things - a network of components with various sensors that communicate with each other [1]
- **MQTT**: Message Queuing Telemetry Transport - a standardized method of machine-to-machine communication [1]
- **PII**: Personally Identifiable Information - any information that can be used to identify, contact or locate a single person [2]
- **SCEMAS**: Smart City Environmental Monitoring & Alert System

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1.5 Overview

Section 2 provides an overall description of the product, including its context, major functions, user characteristics, constraints, and assumptions. Section 3 presents the system use case diagram. Section 4 details the functional requirements in the form of use cases and global scenarios. Section 5 specifies the non-functional requirements, including Look and Feel Requirements, Usability and Humanity Requirements, Performance Requirements, Operational and Environmental Requirements, Maintainability and Support Requirements, Security Requirements, Cultural and Political Requirements and Legal Requirements. Section 6 specifies any future innovations that can be explored. Lastly, Appendix A describes the division of labour among team members.

2 Overall Product Description

2.1 Product Perspective

SCEMAS is going to be developed as a web platform. Related products include other weather applications such as the Weather Network, Apple Weather, and government air quality monitoring platforms. What makes **SCEMAS** different is the focus on providing real-time updates and having a unified dashboard for management.

This product will allow city operators to view system health and active alerts, access historical data, and view real time visualizations. This product will highlight critical alerts to allow city operators to see which issues are a priority in being addressed.

SCEMAS will also interact with real-world mapping software, through Google Maps or Apple Maps APIs to show geographical maps of sensor locations. This product will also utilize the current authentication interfaces to enforce a role-authenticated dashboard, and a rule-based alerting engine to notify subscribed users about critical environmental data.

SCEMAS will also interface with physical hardware sensors, and as such will use an **MQTT** protocol to ensure standardization of incoming sensor telemetry.

Overall, this product serves as an intelligent alerting subsystem within the larger Smart City architecture. It works by transforming raw sensor data into secure and actionable information that city operators can respond to and citizens can view to stay informed.

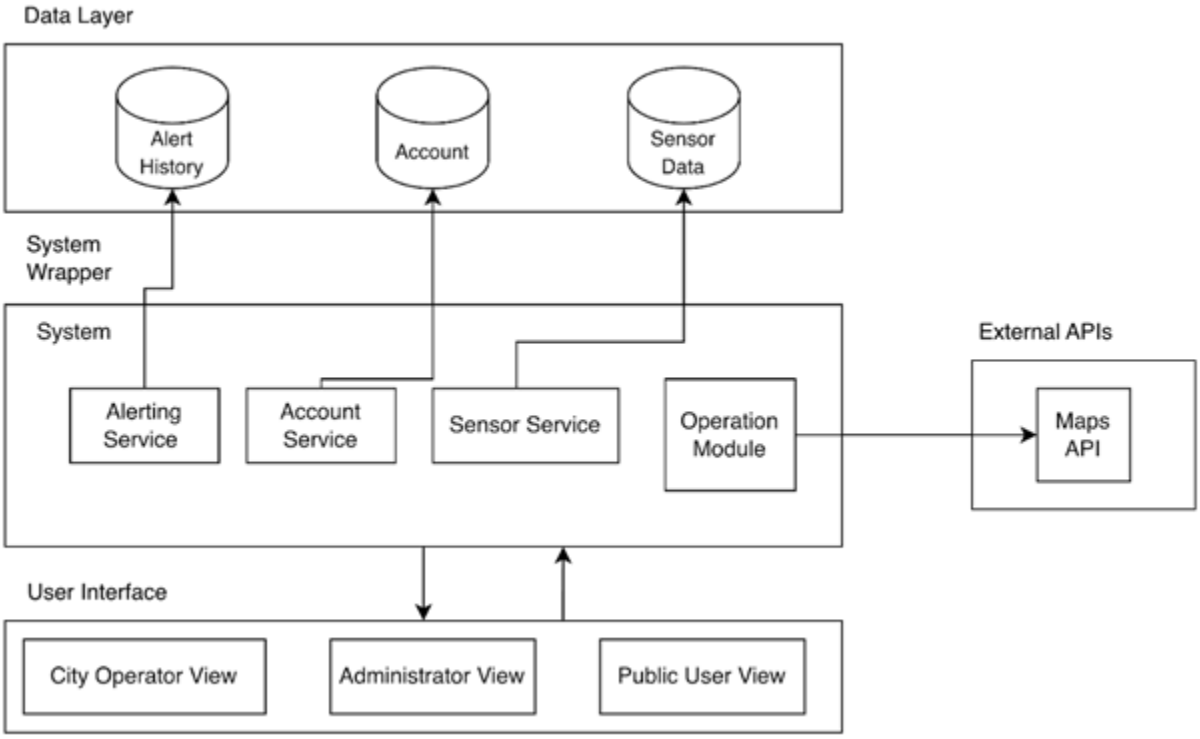

Figure 1: System Diagram

2.2 Product Functions

Modules	Functions
Sensor Service	• Receive sensor data – Allows the system to collect real-time environmental data from connected sensors. • Validate sensor data – Allows sensor data to be checked for correct format, adherence to schema, and plausible values. • Save sensor data – Allows the system to securely store validated sensor data.
Alerting Service	• Subscribe to alerts – Allows users to optionally choose which regions to subscribe to for alerts. • Create alert rules – Allows system administrators to define conditions/criteria that determine when alerts are automatically triggered. • Send alert – Allows the system to notify subscribed users when alerts are triggered. • View critical alerts – Allows city operators to view high-priority alerts in real time. • Change alert status – Allows city operators to update the status of an alert (e.g., active, acknowledged, resolved).

Modules	Functions

Operational Module	• View system health
	  – Allows city operators to monitor overall system performance, availability, and operational status.
	• View historical sensor data
	  – Allows city operators to view log history of past environmental alerts and their status.
	• View audit log
	  – Allows city operators to review a history of system activities and user actions.
	• View data visualizations
	  – Allows city operators to view graphical representations of sensor data.

Account Service	• Create account
	  – Allows users to create accounts.
	• Assign role
	  – Allows users to be assigned a role to make their account compatible with the RBAC system.
	• Login
	  – Allows users to login/logout of their account.

Public Module	• Select region
	  – Allows users to select a region of current interest.
	• Present environmental indicators
	  – Allows users to view information about current environmental indicators for selected regions.

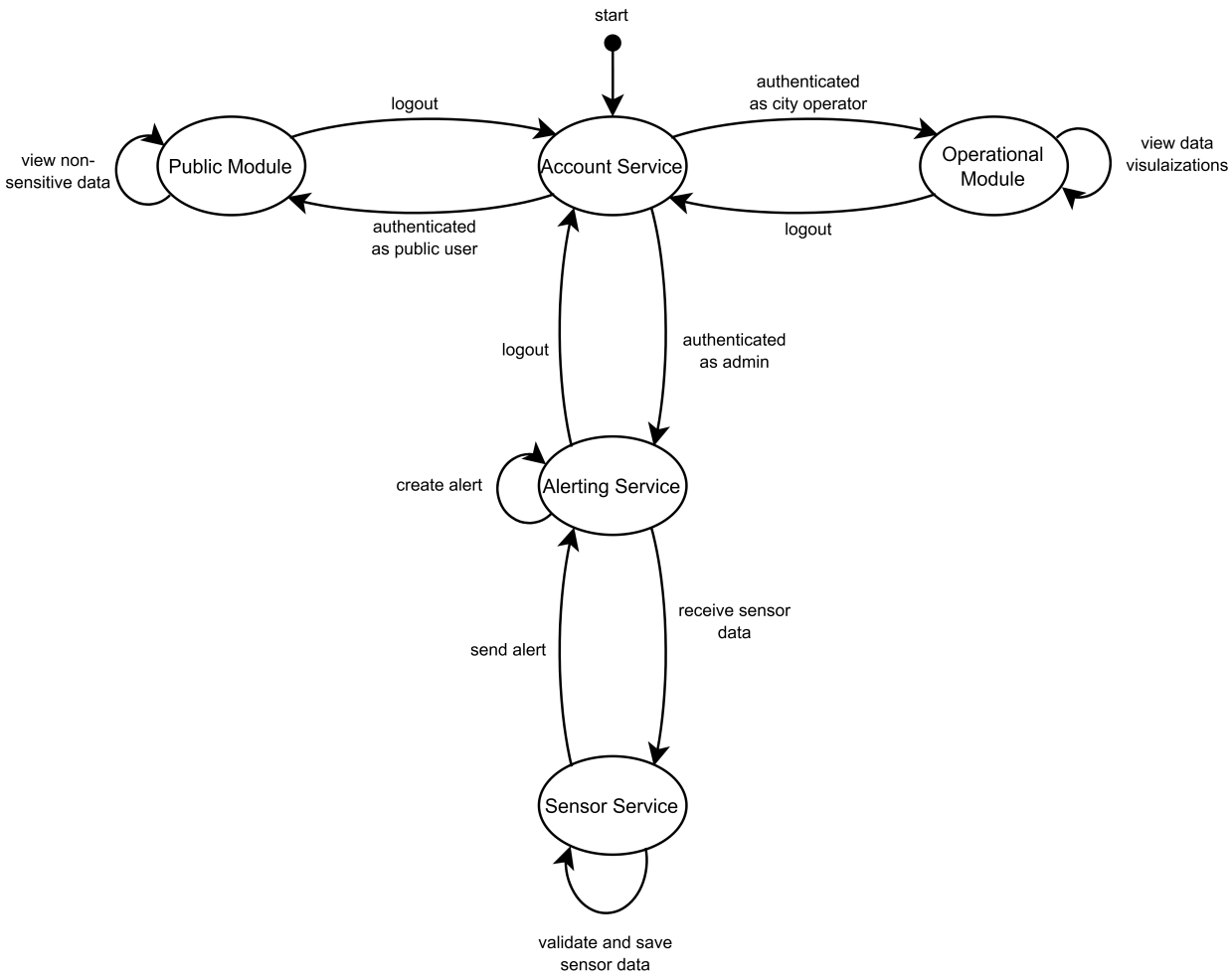

Figure 2: State Diagram

2.3 User Characteristics

This application is designed to be easy for anyone to use and is expected to meet the following criteria:

1. **Education Level:** Basic literacy
   - A person with basic reading, writing, listening, and speaking skills should be able to use the app.
2. **Experience**
   - For public users, no experience is required. Public users should find it easy to navigate through the application.
   - City operators and administrators require sufficient training to operate the application via API to avoid any systematic issues.
3. **Technical Expertise:** Electronic device usage
   - A person with basic familiarity with using an electronic device should be able to utilize the application.

2.4 Constraints

1. **Time constraint:** The time available for the development of this project determines the overall size of the application and the complexity of the system.
2. **Budget constraint:** The project budget limits what tools, resources and technologies developers are able to use. This can affect the functionality and scale of the system.

2.5 Assumptions and Dependencies

- **Hardware:** Physical sensors have already been developed and are able to be installed throughout the city to allow for monitoring the entire area.
- **Human response:** There must be trained personnel dedicated to review and react accordingly to the incoming data.
- **Database/Servers:** It is assumed that there will be servers large enough to sufficiently store and allow easy access to long-term sensor data.
- **Network Connection:** There will be a consistent network connection to allow for sensors to constantly communicate with the system.
- **Citizen Accessibility:** The majority of citizens will have some type of device which can access the internet to view public data.
- **Libraries/Frameworks:** The system will depend on various libraries for JSON data, data visualisation, and authentication services.

2.6 Apportioning of Requirements

Certain requirements are not essential for the primary function of the system and must be delayed for future versions to allow the system to begin running as soon as possible.

1. **Citizen Reports:** Citizens of the city will be able to report certain areas if they believe the sensors and reports are inaccurate compared to their experience of the area.
2. **Language Availability:** Future versions shall provide multiple language options to provide maximum accessibility for citizens.
3. **Trend Prediction (Innovative Feature):** Models will analyse long-term data to predict future trends in the environment and allow for proactive responses.
4. **Event Scheduling:** Authorized personnel will be able to schedule events that will, or have occurred in the city to improve the trend prediction model.
5. **Enhanced Security:** Security of the various components of the system is important, but enhanced security can be pushed to future versions.

3 Use Case Diagram

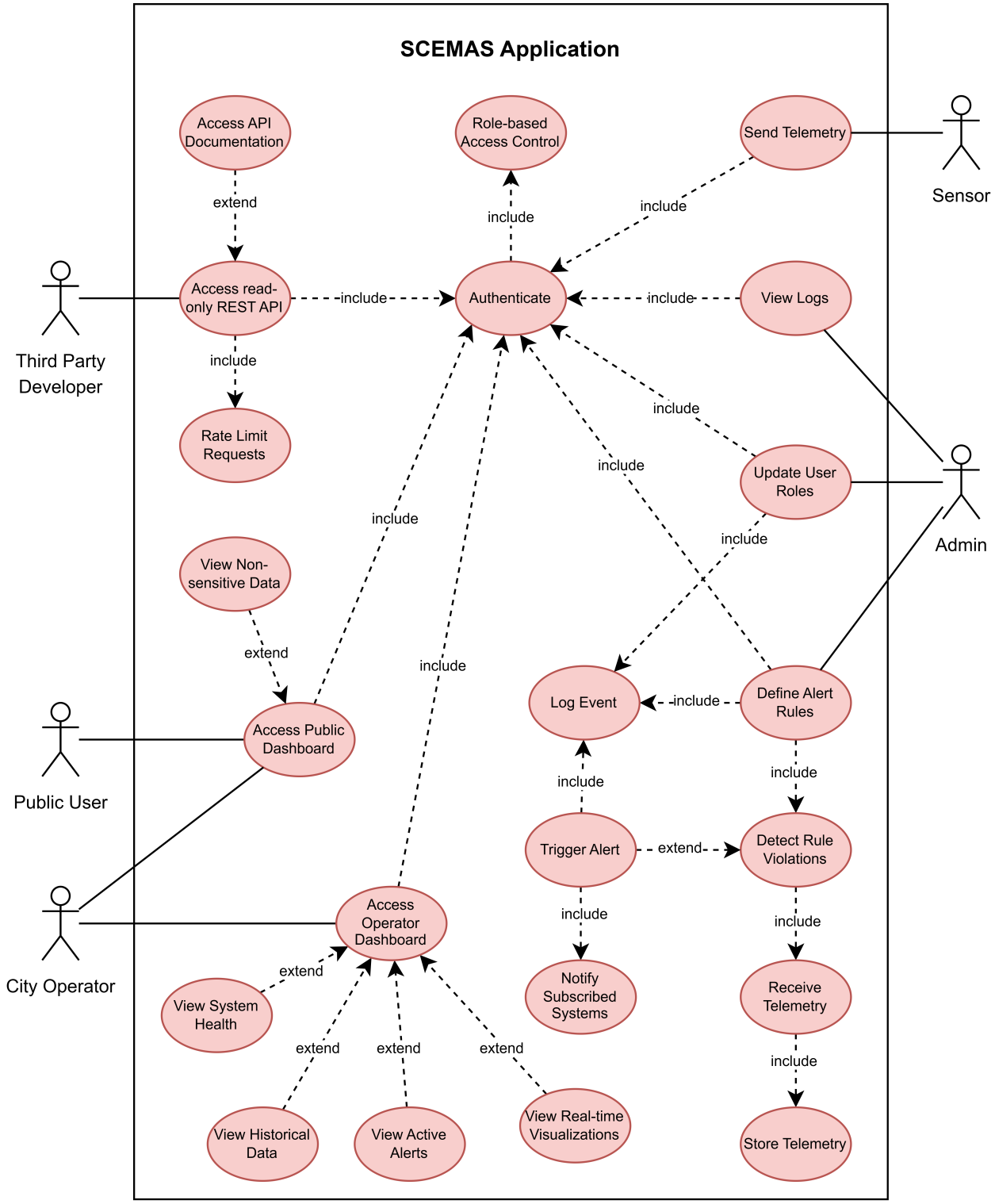

Figure 3: Use Case Diagram

4 Highlights of Functional Requirements

Main Business Events:

- BE1. Define an Alert Rule
- BE2. Create an Account
- BE3. Set an Account Role
- BE4. Login
- BE5. Subscribe to Alerts
- BE6. Manage Alert Status
- BE7. View Historical Sensor Data
- BE8. View Audit Logs
- BE9. View Real-Time Environmental Data

Viewpoints:

- VP1. Public User
- VP2. System Administrator
- VP3. Third Party Developer
- VP4. City Operator
- VP5. Customer Support

BE1. Define an Alert Rule #1

Pre-condition: The user must have a valid administrator account.

VP1. Public User #1
N/A

VP2. System Administrator #2

Main Success Scenario

1. Administrator opens the **SCEMAS** web dashboard.
2. System requires authentication and displays login fields.
3. Administrator enters valid credentials.
4. System authenticates the administrator and displays the administrative dashboard.
5. Administrator navigates to the Alert Rules Management section.
6. System displays a list of existing alert rules and an option to create a new rule.
7. Administrator selects "Create New Rule".
8. System displays a rule configuration form prompting for:
   i. Environmental metric
   ii. Threshold value
   iii. Geographical zone
   iv. Severity level
9. Administrator enters valid rule parameters and submits the form.
10. System validates the rule configuration.
11. System stores the rule and activates it.
12. System displays a confirmation message indicating the rule was successfully created and activated.

Secondary Scenario

4i. System fails to authenticate the administrator.

4i.1 System detects invalid or expired administrator credentials.
4i.2 System displays an authentication error message and prompts to retry login.
9i. Administrator enters invalid or incomplete rule parameters.
9i.1 System detects invalid input.
9i.2 System displays validation errors and prompts correction.
10i. System fails to validate the rule configuration due to internal error.
10i.1 System logs the failure.
10i.2 Define an Alert Rule failed.
11i. System fails to store or activate the rule.
11i.1 System logs the error.
11i.2 Define an Alert Rule failed.

VP3. Third Party Developer #3
N/A

VP4. City Operator #4
N/A

VP5. Customer Support #5

4i. System should display an authentication error message with a link to customer support.
9i. System should display a contextual error message with a link to customer support.
10i. System should log the validation failure and allow customer support to access diagnostic information.
11i. System should log the rule creation or activation failure and provide customer support access to error details.

Global Scenario:

Pre-condition: The user must have a valid administrator account.

Main Success Scenario

1. Administrator opens the **SCEMAS** web dashboard.
2. System requires authentication and displays login fields.
3. Administrator enters valid credentials.
4. System authenticates the administrator and displays the administrative dashboard.
5. Administrator navigates to the Alert Rules Management section.
6. System displays a list of existing alert rules and an option to create a new rule.
7. Administrator selects "Create New Rule".
8. System displays a rule configuration form prompting for:
   i. Environmental metric
   ii. Threshold value
   iii. Geographical zone
   iv. Severity level
9. Administrator enters valid rule parameters and submits the form.
10. System validates the rule configuration.
11. System stores the rule and activates it.
12. System displays a confirmation message indicating the rule was successfully created and activated.

Secondary Scenario

4i. System fails to authenticate the administrator.

4i.1 System detects invalid or expired administrator credentials.

4i.2 System displays an authentication error message and prompts retry or connection to customer support.

4i.3 User re-enters credentials.

4i.4 Return to BE1.4.

Secondary:

4i.3i.1 User chooses to contact customer support.

4i.3i.2 System displays the customer support page.

9i. Administrator enters invalid or incomplete rule parameters.

9i.1 System detects invalid input.

9i.2 System displays validation errors and prompts correction.

9i.3 Return to BE1.9.

10i. System fails to validate the rule configuration due to internal error.

10i.1 System logs the failure.

10i.2 System prompts the administrator to retry or contact customer support.

10i.3 User chooses to retry.

10i.4 Return to BE1.8.

Secondary:

10i.3i.1 User chooses to contact customer support.

10i.3i.2 System displays the customer support page.

11i. System fails to store or activate the rule.

11i.1 System logs the error.

11i.2 System prompts the administrator to contact customer support.

11i.3 User chooses to contact customer support.

11i.4 System displays the customer support page.

BE2. Create an Account #2

Pre-condition: User does not have an existing account and there exists at least one system administrator.

VP1. Public User #1

Main Success Scenario

1. User opens the **SCEMAS** web dashboard.
2. User selects "Create Account".
3. System displays a form requesting:
   i. Full name
   ii. Email address
   iii. Password
4. User enters valid information and submits the form.
5. System validates the information.
6. System creates the user account.
7. System displays a confirmation message indicating successful account creation.
8. System assigns a default Standard role to the user.

Secondary Scenario

4i. User enters invalid or incomplete input.

4i.1 System detects invalid or incomplete user input.
4i.2 System displays validation errors and prompts to retry.
6i. System fails to create the account due to internal error.
6i.1 System logs the error.
6i.2 Create an Account failed.

VP2. System Administrator #2

8i. An existing system administrator updates the user's role to a System Administrator.

VP3. Third Party Developer #3

9i. User generates API keys to use the read-only REST API.

VP4. City Operator #4

8i. An existing system administrator updates the user's role to a City Operator.

VP5. Customer Support #5

4i. System should display an authentication error message with a link to customer support.
6i. System should display a contextual error message with a link to customer support.

Global Scenario:

Pre-condition: User does not have an existing account and there exists at least one system administrator.

Main Success Scenario

1. User opens the **SCEMAS** web dashboard.
2. User selects "Create Account".
3. System displays a form requesting:
   i. Full name
   ii. Email address
   iii. Password
4. User enters valid information and submits the form.
5. System validates the information.
6. System creates the user account.
7. System displays a confirmation message indicating successful account creation.
8. System assigns a default Standard role to the user.

Secondary Scenario

4i. User enters invalid or incomplete input.
4i.1 System detects invalid or incomplete user input.
4i.2 System prompts the user to retry or contact customer support.
4i.3 User chooses to retry.
4i.4 Return to BE2.4.
Secondary:
4i.3i.1 User chooses to contact customer support.
4i.3i.2 System displays the customer support page.

6i. System fails to create the account due to internal error.
6i.1 System logs the error.
6i.2 System prompts the administrator to contact customer support.

6i.3 User chooses to contact customer support.

6i.4 System displays the customer support page.

8i. User wants to create an administrator account.

8i.1 An existing system administrator updates the user's role to a System Administrator.

8ii. User wants to create an operator account.

8ii.1 An existing system administrator updates the user's role to a City Operator.

9i. User is a third party developer.

9i.1 User generates API keys to use the read-only REST API.

BE3. Set an Account Role #3

Pre-condition: There exists at least one system administrator and one other account.

VP1. Public User #1

13ii. User's account is updated to a Public User.

VP2. System Administrator #2

Main Success Scenario

1. Administrator opens the **SCEMAS** web dashboard.
2. System requires authentication and displays login fields.
3. Administrator enters valid credentials.
4. System authenticates the administrator and displays the administrative dashboard.
5. Administrator navigates to the User Management section.
6. System displays a search bar and filters.
7. Administrator searches for a user account.
8. Administrator selects to update the account role.
9. System displays a list of the following roles for the administrator to choose from:
   i. Public User
   ii. System Administrator
   iii. Developer
   iv. City Operator
   v. Customer Support
10. Administrator selects a role.
11. System prompts the administrator to confirm it.
12. Administrator confirms.
13. System activates role change and stores it in the audit log.
14. System displays a confirmation message indicating the role change was successful.

Secondary Scenario

4i. System fails to authenticate the administrator.

4i.1 System detects invalid or expired administrator credentials.

4i.2 System displays an authentication error message and prompts to retry login.

7i. Administrator enters an invalid search.

7i.1 System detects invalid input.

7i.2 System displays validation errors and prompts correction.

7ii. Search returns no results.

7ii.1 System finds no accounts matching the search.

7ii.2 System displays a message indicating no valid results were found.

10i. Administrator selects a role already assigned to the account.
  10i.1 System prompts administrator to select a different role.
13i. System fails to change roles.
  13i.1 System cannot change role of the account.
  13i.2 System displays message to administrator indicating failure.

VP3. Third Party Developer #3

13iii. User's account is updated to a Developer.

VP4. City Operator #4

13iv. User's account is updated to a City Operator.

VP5. Customer Support #5

13v. User's account is updated to Customer Support.

Global Scenario:

Pre-condition: There exists at least one system administrator and one other account.

Main Success Scenario

1. Administrator opens the **SCEMAS** web dashboard.
2. System requires authentication and displays login fields.
3. Administrator enters valid credentials.
4. System authenticates the administrator and displays the administrative dashboard.
5. Administrator navigates to the User Management section.
6. System displays a search bar and filters.
7. Administrator searches for a user account.
8. Administrator selects to update the account role.
9. System displays a list of the following roles for the administrator to choose from:
   i. Public User
   ii. System Administrator
   iii. Developer
   iv. City Operator
   v. Customer Support
10. Administrator selects a role.
11. System prompts the administrator to confirm it.
12. Administrator confirms.
13. System activates role change and stores it in the audit log.
   i. User's account is updated to Public User.
   ii. User's account is updated to Developer.
   iii. User's account is updated to City Operator.
   iv. User's account is updated to Customer Support.
14. System displays a confirmation message indicating the role change was successful.

Secondary Scenario

4i. System fails to authenticate the administrator.
  4i.1 System detects invalid or expired administrator credentials.
  4i.2 System displays an authentication error message and prompts to retry login.

7i. Administrator enters an invalid search.

7i.1 System detects invalid input.

7i.2 System displays validation errors and prompts correction.

7ii. Search returns no results.

7ii.1 System finds no accounts matching the search.

7ii.2 System displays a message indicating no valid results were found.

10i. Administrator selects a role already assigned to the account.

10i.1 System prompts administrator to select a different role.

13i. System fails to change roles.

13i.1 System cannot change role of the account.

13i.2 System displays message to administrator indicating failure.

BE4. Login #4

Pre-condition: User has an existing account.

VP1. Public User #1

Main Success Scenario

1. User opens the **SCEMAS** web application.
2. System displays a login page.
3. User enters valid login credentials.
4. System validates the credentials.
5. System approves the user.
6. System displays the user dashboard.

Secondary Scenario

3i. User enters invalid login credentials.

3i.1 System detects invalid credentials.

3i.2 System displays error message and prompts retry.

4i. System fails to validate credentials due to system error.

4i.1 System displays a message asking the user if they would like to unsubscribe.

4i.2 System logs the error.

4i.3 System prompts user to try again later or contact customer support.

VP2. System Administrator #2

6i. System displays admin dashboard.

VP3. Third Party Developer #3

3i. Developer logs in through API.

VP4. City Operator #4

N/A

VP5. Customer Support #5

N/A

Global Scenario:

Pre-condition: User has an existing account.

Main Success Scenario

1. User opens the **SCEMAS** web application.

2. System displays a login page.
3. User enters valid login credentials.
4. System validates the credentials.
5. System approves the user.
6. System displays the user dashboard.

Secondary Scenario

3i. User enters invalid login credentials.
  3i.1 System detects invalid credentials.
  3i.2 System displays error message and prompts retry.
3ii. Developer logs in through API.
4i. System fails to validate credentials due to system error.
  4i.1 System displays a message asking the user if they would like to unsubscribe.
  4i.2 System logs the error.
  4i.3 System prompts user to try again later or contact customer support.
6i. System displays admin dashboard.

BE5. Subscribe to Alerts #5

Pre-condition: The user must have an existing account.

VP1. Public User #1

Main Success Scenario

1. User opens the **SCEMAS** web application.
2. System requires the user to log in and displays a login page.
3. User logs in.
4. System validates user login request.
5. System displays **SCEMAS** homepage to user.
6. User chooses to view geographical map locations.
7. User clicks on a specific location.
8. System displays location environmental metrics and relevant buttons.
9. User presses the subscribe button.

Secondary Scenario

6i. Map not available.
  6i.1 System logs the error.
  6i.2 System prompts the user to contact customer support.
  6i.3 User chooses to contact customer support.
  6i.4 System displays the customer support page.
9i. User already subscribed.
  9i.1 System displays a message asking the user if they would like to unsubscribe.
  9i.2 User submits "No".
  9i.3 System returns to homepage.

VP2. System Administrator #2

N/A

VP3. Third Party Developer #3

Pre-condition: User must be authenticated.

Main Success Scenario

1. User picks a region for subscription.
2. User subscribes to the subscription API.

VP4. City Operator #4
N/A

VP5. Customer Support #5
N/A

Global Scenario:

Pre-condition: User is authenticated with **SCEMAS**.

Main Success Scenario

1. User opens the **SCEMAS** web application.
2. System requires the user to log in and displays a login page.
3. User logs in.
4. System validates user login request.
5. System displays **SCEMAS** homepage to user.
6. User chooses to view geographical map locations.
7. User clicks on a specific location.
8. System displays location environmental metrics and relevant buttons.
9. User presses the subscribe button.

Secondary Scenario

6i. Map not available.
- 6i.1 System logs the error.
- 6i.2 System prompts the user to contact customer support.
- 6i.3 User chooses to contact customer support.
- 6i.4 System displays the customer support page.

9i. User already subscribed.
- 9i.1 System displays a message asking the user if they would like to unsubscribe.
- 9i.2 User submits "No".
- 9i.3 System returns to homepage.

BE6. Manage Alert Status #6

VP1. Public User #1
N/A

VP2. System Administrator #2
Pre-condition: User is a system administrator with an existing **SCEMAS** account.

Main Success Scenario

1. User opens **SCEMAS** web application.
2. System requires the user to log in and displays a login page.
3. User logs in.
4. System validates user login request.
5. System displays **SCEMAS** dashboard.
6. User navigates to the alerts tab.

7. System displays list of all alerts.
8. User selects an alert from the list of displayed alerts.
9. System displays alert details, including alert status.
10. User deletes alert.
11. System displays a prompt asking the administrator to confirm deletion.
12. User selects "Yes".
13. System deletes selected alert and returns to alert list.

Secondary Scenario

4i. System fails to authenticate the administrator.
  4i.1 System detects invalid or expired administrator credentials.
  4i.2 System displays an authentication error message and prompts retry or connection to customer support.

7i. No alerts available.
  7i.1 System displays error message that no alerts are currently available.
  7i.2 Administrator can return to homepage.

12i. System administrator selects "No".
  12i.1 System displays message that alert was not deleted.
  12i.2 Administrator returns to alert list.

VP3. Third Party Developer #3
N/A

VP4. City Operator #4
Pre-condition: User is a city operator with an existing **SCEMAS** account.

Main Success Scenario

1. User opens **SCEMAS** web application.
2. User logs into the system.
3. System authenticates city operator's account.
4. System displays the **SCEMAS** dashboard.
5. User navigates to the alerts tab.
6. System displays list of all alerts.
7. User selects an alert from the list of displayed alerts.
8. System displays alert details, including alert status.
9. User updates alert status to Active, Acknowledged, or Resolved.
10. User submits alert status update.
11. System logs the update with corresponding timestamp.
12. System displays message to city operator indicating a successful update.
13. User clicks "OK" to acknowledge the successful update.
14. System displays updated alert status.

Secondary Scenario

3i. System fails to authenticate the city operator.
  3i.1 System detects invalid or expired operator credentials.
  3i.2 System displays an authentication error message and prompts retry or connection to customer support.

6i. No alerts available.
  6i.1 System displays error message that no alerts are currently available.
  6i.2 City operator can return to homepage.

VP5. Customer Support #5
N/A

Global Scenario:

Pre-condition: User has an account with **SCEMAS** and is a city operator or administrator.

Main Success Scenario

1. User opens **SCEMAS** web application.
2. User logs into the system.
3. System authenticates the user account.
4. System displays the **SCEMAS** dashboard.
5. User navigates to the alerts tab.
6. System displays list of all alerts.
7. User selects an alert from the list of displayed alerts.
8. System displays alert details, including alert status.
9. User updates alert status to Active, Acknowledged, or Resolved.
10. User submits alert status update.
11. System logs the update with corresponding timestamp.
12. System displays message indicating a successful update.
13. User selects "OK" to acknowledge the successful update.
14. System displays updated alert status.

Secondary Scenario

6i. No alerts available.
  6i.1 System displays error message that no alerts are currently available.
  6i.2 User can return to homepage.

9i. A designated system administrator deletes the alert.
  9i.1 System displays message confirming administrator wants to delete alert.
  9i.2 Administrator confirms deletion.

BE7. View Historical Sensor Data #7

Pre-condition: User is logged in and has access to the application.

VP1. Public User #1

Main Success Scenario

1. User opens **SCEMAS** web application.
2. System displays a login page and requires the user to log in.
3. User logs in.
4. System validates user login request.
5. System displays **SCEMAS** homepage.
6. User selects a geographical region.
7. User selects a time range.
8. System retrieves historical data within the selected time range.
9. System displays historical data using tables and graphs.

Secondary Scenario

2i. User enters invalid login credentials.
  2i.1 System detects invalid credentials.
  2i.2 System displays error message and prompts retry.
8i. System fails to retrieve historical data.
  8i.1 System logs the error.
  8i.2 System displays an error message stating data is unavailable.

VP2. System Administrator #2
N/A

VP3. Third Party Developer #3

6i. System displays admin dashboard.

VP4. City Operator #4
N/A

VP5. Customer Support #5
N/A

Global Scenario:

Pre-condition: User is logged in and has access to the application.

Main Success Scenario

1. User opens **SCEMAS** web application.
2. System displays a login page and requires the user to log in.
3. User logs in.
4. System validates user login request.
5. System displays **SCEMAS** homepage.
6. User selects a geographical region.
7. User selects a time range.
8. System retrieves historical data within the selected time range.
9. System displays historical data using tables and graphs.

Secondary Scenario

2i. User enters invalid login credentials.
  2i.1 System detects invalid credentials.
  2i.2 System displays error message and prompts retry.
8i. System fails to retrieve historical data.
  8i.1 System logs the error.
  8i.2 System displays an error message stating data is unavailable.

BE8. View Audit Log #8

Pre-condition: There exists at least one system administrator.

VP1. Public User #1
N/A

VP2. System Administrator #2
Main Success Scenario

1. Administrator opens the **SCEMAS** web dashboard.

2. System requires authentication and displays login fields.
3. Administrator enters valid credentials.
4. System authenticates the administrator and displays the administrative dashboard.
5. Administrator navigates to the Audit Log section.
6. System displays new audit log access requests.
7. Administrator accepts, rejects, or ignores these requests.
8. System displays audit log as well as filters to search for specific events.
9. Administrator applies filters.
10. System displays updated audit log according to filters.

Secondary Scenario

4i. System fails to authenticate the administrator.
  4i.1 System detects invalid or expired administrator credentials.
  4i.2 System displays an authentication error message and prompts retry login.

9i. Administrator enters an invalid search.
  9i.1 System detects invalid input.
  9i.2 System displays validation errors and prompts correction.

9ii. Search returns no results.
  9ii.1 System finds no records in audit log matching the search.
  9ii.2 System displays a message indicating no valid results were found.

VP3. Third Party Developer #3
N/A

VP4. City Operator #4
N/A

VP5. Customer Support #5

6i. Customer support requests access to sections of the audit log.

Global Scenario:

Pre-condition: There exists at least one system administrator.

Main Success Scenario

1. Administrator opens the **SCEMAS** web dashboard.
2. System requires authentication and displays login fields.
3. Administrator enters valid credentials.
4. System authenticates the administrator and displays the administrative dashboard.
5. Administrator navigates to the Audit Log section.
6. System displays new audit log access requests.
7. Administrator accepts, rejects, or ignores these requests.
8. System displays audit log as well as filters to search for specific events.
9. Administrator applies filters.
10. System displays updated audit log according to filters.

Secondary Scenario

4i. System fails to authenticate the administrator.
  4i.1 System detects invalid or expired administrator credentials.
  4i.2 System displays an authentication error message and prompts retry login.

9i. Administrator enters an invalid search.

9i.1 System detects invalid input.

9i.2 System displays validation errors and prompts correction.

9ii. Search returns no results.

9ii.1 System finds no records in audit log matching the search.

9ii.2 System displays a message indicating no valid results were found.

BE9. View Real-Time Environmental Data #9

Pre-condition: The **SCEMAS** system is operational, and sensor telemetry is being ingested, validated, and stored. Recent aggregated environmental data is available (or cached), and the user has appropriate access to a **SCEMAS** interface.

VP1. Public User #1

Main Success Scenario

1. User accesses the **SCEMAS** public dashboard.
2. System displays a map of predefined city zones.
3. System shows aggregated real-time environmental metrics (e.g., AQI, temperature, noise).
4. User selects a zone or metric.
5. System updates the visualization with summary values and near real-time charts.
6. User optionally views recent historical trends (e.g., last hour averages).

Secondary Scenario

3i. Data for the selected zone is unavailable.

3i.1 System displays the most recent cached data or an informative message.

3i.2 System indicates data freshness and suggests selecting another zone.

VP2. System Administrator #2

Main Success Scenario

1. System Administrator logs into the secure **SCEMAS** dashboard.
2. System displays real-time visualizations, raw and aggregated sensor streams, and system health indicators.
3. Administrator verifies data pipeline operation and visualization updates.
4. Administrator confirms data validity and storage.

Secondary Scenario

2i. Visualization service fails to load.

2i.1 System displays an error and logs the incident.

3i. Invalid sensor data is detected.

3i.1 System flags the data and records an audit log entry.

VP3. Third Party Developer #3

Main Success Scenario

1. Developer sends an authenticated request to the **SCEMAS** public REST API.
2. System authorizes the request.
3. System returns real-time aggregated sensor data in structured JSON format.
4. Developer consumes the data in an external application.

Secondary Scenario

2i. API rate limit exceeded.

2i.1 System rejects the request and returns a rate-limit error response.

VP4. City Operator #4

Main Success Scenario

1. City operator logs into the operator dashboard.
2. System displays real-time maps, numerical readings, charts, and gauges.
3. Operator applies filters (zone, metric type, time window).
4. Operator monitors data for anomalies or threshold warnings.
5. Operator uses insights to support operational decisions or escalations.

Secondary Scenario

3i. A visualization or data stream fails to refresh on time.

3i.1 System displays a warning and allows manual refresh.

VP5. Customer Support #5

Main Success Scenario

1. Customer support accesses the system in read-only support mode.
2. System displays summarized real-time visualizations and sensor readings.
3. Support references the data to respond to public or operator inquiries.

Global Scenario:

Pre-condition: The **SCEMAS** system is operational, and sensor telemetry is being ingested, validated, and stored. Recent aggregated environmental data is available (or cached), and the user has appropriate access to a **SCEMAS** interface.

Main Success Scenario

1. User accesses the **SCEMAS** interface or API.
2. System retrieves the latest validated and aggregated sensor data.
3. System renders real-time environmental information appropriate to the user's role:
4. Visual dashboards (maps, charts, gauges, indicators), and/or
5. Numerical values, trends, or structured API responses.
6. User interacts with the data (e.g., selecting zones, metrics, or time windows).
7. User successfully views current environmental conditions.

5 Non-Functional Requirements

5.1 Look and Feel Requirements

5.1.1 Appearance Requirements

LF-AP1. The system dashboard shall use a consistent layout across all pages.

Rationale: Having inconsistent layouts often lead to user frustration and confusion [3]. As such, incorporating a pre-defined layout will increase user engagement and streamline the process for new users as they try to navigate through the pages of the web dashboard.

LF-AP2. The system shall use standardized colour to indicate hazard severity.

Rationale: Standardized risk colours improves hazard recognition speed, enhance risk perception, and help communicate through language/education barriers [4].

LF-AP3. The system dashboard should be able to adapt to fit desktops, phones, and tablets.

Rationale: Keeping this app accessible to people anywhere, anytime is important as well as allowing those with only one of these 3 devices to still be able to access important data.

5.1.2 Style Requirements

LF-ST1. The system shall use a professional and minimalistic visual design.

Rationale: Keeping the tone of the dashboard minimalistic and professional accurately conveys the importance of the data, while also keeping the dashboard as easily readable as possible.

LF-ST2. The system shall use clear fonts with readable font sizes, as well as proper spacing between interface elements.

Rationale: Proper font sizes will allow those with poor eyesight to still be able to use the app as well as using certain fonts helps those with reading disabilities [5]. Keeping it accessible to the most amount of groups of people is important.

LF-ST3. Consistent iconography and terminology shall be used throughout the app.

Rationale: Remaining consistent in icons and terminology will help members of the public, who will likely be new to these topics, to understand the data the dashboard is displaying.

5.2 Usability and Humanity Requirements

5.2.1 Ease of Use Requirements

UH-EOU1. The system shall utilize descriptive buttons and icons unless they are universal and have intuitive meanings.

Rationale: Universal icons include ones that are recurring and have repeated functionalities. The rest of the buttons and icons should not cause confusion amongst users and should be able to guide user action [6].

UH-EOU2. The system must have clear exit points during any process.

Rationale: Users should be able to easily exit a process and go back in the case of accidental click or change in decision making [6].

5.2.2 Personalization and Internationalization Requirements

UH-PI1. The system must support multiple languages and allow users to set their preferred language for app navigation.

Rationale: Users must be able to understand and navigate through the application for the best user experience. This improves long-term accessibility and inclusivity [7].

UH-PI2. The system shall allow users to subscribe to alerts for specific geographical regions.

Rationale: Users primarily care for different areas of cities, such as areas they are situated in. This improves relevancy and increases usability amongst users.

5.2.3 Learning Requirements

UH-L1. First time users shall be able to navigate through the application and perform basic tasks without prior training.

Rationale: The user interface should be inherent enough that first time users spend little time trying to figure out how the application works. If the interface is deemed too complicated it can reduce user retention and disinterest users [6].

5.2.4 Understandability and Politeness Requirements

UH-UP1. All system messages shall be presented using clear and concise language.

Rationale: Users must be able to interpret messages without any confusion, especially for non-technical users. If messages create confusion for users, it may lead to users being unsatisfied with the application [8].

UH-UP2. All system messages shall be polite and should use appropriate phrasing.

Rationale: Any messages presented by the system should not be deemed as rude or inappropriate and should not in any way offend the user. Inappropriate phrasings can lead to user dissatisfaction and can cause major long term issues for the application [9].

5.2.5 Accessibility Requirements

UH-A1. The system shall be designed using colors that are adhered to users with colorblindness.

Rationale: Users who are colorblind should be able to identify different colors in the application without any issues to avoid confusion of different sections [10].

5.3 Performance Requirements

5.3.1 Speed and Latency Requirements

PR-SL1. Generated alerts must be visible on the dashboard within 1 seconds of a threshold breach being identified.

Rationale: An acceptable web response time delay is considered to be between 200 milliseconds to 1 seconds [11]. Since generating alerts is the most time-sensitive component of **SCEMAS**, the alerts must be completed within this acceptable time delay.

PR-SL2. Sensor data must become available within 3 seconds of being validated.

Rationale: Extreme weather can permanently affect individuals within 2 minutes [12]. Having a quick validation time of 3 seconds ensures that even if the operator takes up to 30 seconds to address the alert, and additional time to send out a public announcement, users will get the necessary information to stay safe.

5.3.2 Safety-Critical Requirements

N/A

5.3.3 Precision or Accuracy Requirements

PR-PA1. Environmental readings collected through sensors must be accurate within 5% of the true value.

Rationale: Commercially available sensors are often reported to be accurate with a 5% error margin [13]. It is important to use sensors with such accuracy to maintain public usage, as a correlation between inaccurate data and reduced trust in current competing apps has been identified [14].

PR-PA2. Environmental readings displayed to users should be displayed precisely in reference to collected data within 1 unit.

Rationale: Not displaying environmental factors to decimal places complies with METAR reporting guidelines [15]. This ensures the application is in-line with universal standards and it also ensures a more readable user-friendly interface.

5.3.4 Reliability and Availability Requirements

PR-RA1. The web application must be available 99.95% of the time.

Rationale: Having an application with high availability is crucial in the context of **SCEMAS** to ensure relevant alerts are being delivered in real-time. Having a 99.95% availability aligns with the current industry standards, effectively reducing the risk of operational delays and building user trust in the **SCEMAS** system [16].

PR-RA2. The web application must have a backup of data.

Rationale: Having **SCEMAS** data backed-up will help in getting a the system back up and running in the event of an availability loss [17]. These backups will ensure all critical information such as pending alerts are saved and stored in the case of an outage. These backups can then be used to ensure that critical alerts are still being delivered.

5.3.5 Robustness or Fault-Tolerance Requirements

PR-RFT1. The system must allow city operators to view active alerts even when the dashboard loses internet connection.

Rationale: Allowing city operators to view critical alerts even when connectivity is lost allows **SCEMAS** to quickly recover from the outage, and ensure that delays are prevented when connection is restored.

5.3.6 Capacity Requirements

PR-C1. The system must be able to handle simultaneous input from 1000 sensors.

Rationale: Ensuring that **SCEMAS** is able to respond to 1000 sensors is crucial as various sensors will be required in each pre-determined geographical location. Continuous data will need to be validated by all of these sensors simultaneously to prevent bottlenecking and ensure system responsivity.

PR-C2. Sensor data must be collected at a rate of 100 readings / seconds.

Rationale: With a sample rate lower than 100 readings / seconds, **SCEMAS** will see evident delays in information being reported to the user interface. As such maintaining this sample rate is important for ensuring a continuous and "real-time" environmental indicator.

5.3.7 Scalability or Extensibility Requirements

PR-SE1. The code must follow the SOLID design principles.

Rationale: The SOLID design principles ensure maintainability and flexibility as a system grows, which allows for long-term sustainability [18]. By following SOLID design principles, it is effectively ensured that **SCEMAS** will easily be able to scale or extend in the future given new requirements.

5.3.8 Longevity Requirements

N/A

5.4 Operational and Environmental Requirements

5.4.1 Expected Physical Environment

OE-EPE1. The system shall run on cloud servers instead of local, on-site hardware.

Rationale: Hosting the system in the cloud makes it easier to deploy, scale, and maintain. This setup fits well with smart city and **IoT** use cases, where flexibility and remote access matter [19].

5.4.2 Requirements for Interfacing with Adjacent Systems

OE-IA1. The system shall interface with external **IoT** devices using the **MQTT** protocol over TCP/IP.

Rationale: **MQTT** is widely used in **IoT** systems and works well even in low-bandwidth or unreliable network conditions [20]. Using **MQTT** ensures compatibility with common sensor platforms.

OE-IA2. The system shall expose a RESTful API to allow external applications to retrieve aggregated, non-sensitive environmental data.

Rationale: A REST API is a standard way for other systems to access data, making it easier to connect the system to public displays, dashboards, or third-party apps [21].

OE-IA3. The system shall integrate with a third-party mapping service (e.g., Google Maps or Apple Maps APIs) to display sensor locations and geographic data.

Rationale: Maps make it much easier for city operators to see where environmental conditions are happening. Using existing mapping services also saves development time and provides reliable location features.

5.4.3 Productization Requirements

N/A

5.4.4 Release Requirements

OE-R1. Each system release shall include versioned API endpoints.

Rationale: API versioning prevents breaking changes for existing users of the public API and allows the system to evolve over time [21].

OE-R2. The system shall be accessible through modern web browsers (Chrome, Firefox, Edge, Safari) released within the last five years.

Rationale: Supporting modern browsers helps ensure a smooth user experience while keeping development and testing manageable.

5.5 Maintainability and Support Requirements

5.5.1 Maintenance Requirements

MS-M1. The system shall log system events, errors, and alerts in a centralized logging mechanism.

Rationale: Centralized logging allows developers and administrators to monitor system behavior, diagnose issues, and support ongoing maintenance.

MS-M2. The system shall allow users to report software bugs or system issues through the web dashboard.

Rationale: Built-in bug reporting makes it quicker and easier for users to report problems accurately, without needing external tools.

5.5.2 Supportability Requirements

MS-S1. The public REST API shall be accompanied by developer documentation describing endpoints, parameters, authentication requirements, and response formats.

Rationale: API documentation helps third-party developers understand how to access and use public data without requiring direct support [21].

MS-S2. The system shall provide user documentation for the operational dashboard, covering core functions such as alert acknowledgment, data visualization, and system health monitoring.

Rationale: Most users won't know how the system works internally. Clear documentation helps them use the dashboard correctly and confidently.

5.5.3 Adaptability Requirements

MS-A1. The system shall allow new sensor types and environmental metrics to be added without requiring major architectural changes.

Rationale: Supporting additional sensor types allows the system to adapt to future monitoring needs and remain useful as city requirements change.

5.6 Security Requirements

5.6.1 Access Requirements

SR-AC1. The system shall restrict access to administrative dashboards and management functionality to authenticated users only.

Rationale: If all users were to have management functionality, there would not be any data protected, and all sensor data would be available to all users. This is to follow the Bell-LaPadula Security Model [22].

SR-AC2. The system shall enforce role-based access control such that users may only access features and data permitted by their assigned role (e.g., public users shall have read-only access).

Rationale: If public users were to have editing access, there is a security issue, as some users might change data, which destroys the integrity of the sensors. This is to follow the Bell-LaPadula Security Model [22].

SR-AC3. Third-party access to system data shall be permitted only through authenticated and authorized API endpoints.

Rationale: Supporting additional sensor types allows the system to adapt to future monitoring needs and remain useful as city requirements change. This is to follow the Bell-LaPadula Security Model [22].

5.6.2 Integrity Requirements

SR-INT1. The system shall validate incoming sensor data to ensure it conforms to expected formats and value ranges before processing or storage.

Rationale: If sensor data is not validated, then misinformation may be conveyed to the user. This would violate the Bell-LaPadula model [22].

SR-INT2. The system shall protect data in transit between system components and client interfaces from unauthorized modification.

Rationale: Data encryption is necessary to protect the data of the users from being stolen. Common systems that use sensor data such as Ring camera doorbells encrypt their data as well [23].

SR-INT3. The system shall prevent unauthorized alteration of stored environmental data.

Rationale: If public users were to have editing access, there is a security issue, as some users might change data, which destroys the integrity of the sensors. This would violate the Bell-LaPadula model [22].

5.6.3 Privacy Requirements

SR-P1. The system shall not collect, store, or process personally identifiable information, unless disclosed in a privacy policy and obtained with user consent.

Rationale: Each user is entitled to the privacy of their own sensor data, which can create a distrust if that data is collected in any way. This is also required for our product to be implemented in Canada [24].

SR-P2. Environmental data exposed to public users and third-party developers shall be aggregated and non-sensitive.

Rationale: Maintain security framework, as general users do not need access to sensitive data [25].

SR-P3. The system shall limit access to detailed or raw data to authorized internal roles only.

Rationale: Raw data is very sensitive, and if altered can change sensor data.

5.6.4 Audit Requirements

SR-AU1. The system shall record audit logs for authentication and authorization events.

Rationale: This is to gather information on all events that access sensitive data, ensuring that there is record of when this data is accessed.

SR-AU2. The system shall log administrative actions that affect system configuration or access control.

Rationale: This is to gather information on all events that access sensitive data, ensuring that there is record of when this data is accessed.

SR-AU3. Audit logs shall support system monitoring, troubleshooting, and accountability.

Rationale: This is to gather information on how the users use the app and to fix any potential bugs that users might face during their usage.

5.6.5 Immunity Requirements

SR-IM1. The system shall implement basic protection mechanisms (e.g., input validation) to reduce exposure to malformed or malicious requests.

Rationale: This is to ensure that there is protection against users who might be using the software for malicious intent.

SR-IM2. The system shall apply rate limiting or throttling on public APIs to mitigate abuse or denial-of-service attempts.

Rationale: This is to keep the data protected, and not have the data accessed by brute force.

SR-IM3. The system shall isolate failures in individual components to reduce impact on overall system availability.

Rationale: This is to centralize problems to make troubleshooting easier.

5.7 Cultural and Political Requirements

5.7.1 Cultural Requirements

CP-CU1. The system shall support various languages and measurement units.

Rationale: Smart city systems must support diverse populations and allow municipalities to configure region-specific preferences.

CP-CU2. The system shall avoid using culturally significant symbols or terminology.

Rationale: Neutral terminology ensures information is universally understandable and avoids cultural misinterpretation.

5.7.2 Political Requirements

CP-PO1. The system shall present all environmental data accurately without political bias.

Rationale: Environmental data has been shown to be able to influence public concern and impact political priorities as well as voter choices [26], so the data we present must be as objective as possible.

CP-PO2. The system will work closely with local municipalities.

Rationale: Local municipalities are responsible for enforcing environmental regulations and reacting to public safety risks. Working closely with our system enables them to do this more effectively.

5.8 Legal Requirements

5.8.1 Compliance Requirements

LR-COMP1. All personal information collected (including personal information given to third parties for processing purposes) must be protected.

Rationale: This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 1 - Accountability [27].

LR-COMP2. The purposes for all personal information being collected (including personal information given to third parties for processing purposes) must be identified before or at the time it is being collected.

Rationale: This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 1 - Accountability [27].

LR-COMP3. The knowledge and consent of the user for all personal information collected (including personal information given to third parties for processing purposes) is required.

Rationale: This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 1 - Accountability [27].

5.8.2 Standards Requirements

LR-STD1. The app should not use notifications for promotional/advertisement purposes.

Rationale: This is a standard for app quality in terms of visual aspect by Google [28].

LR-STD2. Touch targets in the app should have a size of at least 48dp.

Rationale: This is a standard for app quality in terms of visual aspect by Google [28].

LR-STD3. Any personal or sensitive data should not be logged to a system log or a log specific to the app.

Rationale: This is a standard for app quality in terms of visual aspect by Google [28].

LR-STD4. The app's internal storage should be used to store any sensitive data.

Rationale: This is a standard for app quality in terms of visual aspect by Google [28].

6 Innovative Feature

The innovative feature selected for implementation is the prediction of future environmental conditions based on historical data trends. This feature extends the system beyond its core functions of monitoring and alerting. Its purpose is to make the system more useful for city operators and the public by supporting better understanding and decision-making. Seeing future trends can help city operators plan ahead and allocate support more effectively.

Several other innovative features were also considered:

- Automatic classification of alert severity
- Allowing citizens to submit environmental reports
- Comparing environmental data across different city zones
- Providing an overall environmental "health score" for the city
- Integration with emergency services for high-risk alerts
- Automatic detection of unusual patterns in sensor data

These features were considered as ways to improve the system and make it more proactive, interactive, and useful in real-world smart city scenarios.

A Division of Labour

Division of Labour sheet which indicates the contributions of each team member.

A.1 Bilal, Mohammad

- Section 3: Use Case Diagram
- Section 4: BE1 and BE2
- Section 5: 5.4 and 5.5
- Section 6: Write up
- Document formatting and review
- Section 2.2 Revision: Updated state diagram for second submission
- Ported the entire document to LaTeX for second submission

A.2 Jessani, Kaiden

- Section 1: Introduction
- Section 4: BE9
- Section 5: 5.6 and 5.8
- Section 6: Brainstormed Ideas
- Document formatting and review
- Section 1 Revision: Updated to paragraph format for second submission

A.3 Ludwig, Christopher

- Section 2: Assumptions and Dependencies, Apportioning of Requirements
- Section 4: BE3, BE8
- Section 5: 5.1, 5.7

- Section 6: Brainstormed ideas
- Document formatting and review

A.4 Shahid, Irfan

- Section 2: User Characteristics, Constraints
- Section 4: BE4, BE7
- Section 5: 5.2
- Section 6: Brainstormed ideas
- Document formatting and review

A.5 Patel, Prutha

- Section 2: Productive Perspective, Product Functions
- Section 4: BE5, BE6
- Section 5: 5.3
- Section 6: Brainstormed ideas
- Document formatting and review